



Unsupervised pathology detection in medical images using conditional variational autoencoders

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Abstract

Purpose Pathology detection in medical image data is an important but a rather complicated task. In particular, the big variability of the pathologies is a challenge to automatic detection methods and even to machine learning methods. Supervised algorithms would usually learn the appearance of a single pathological structure based on a large annotated dataset. As such data is not usually available, especially in large amounts, in this work we pursue a different unsupervised approach.

Methods Our method is based on learning the entire variability of healthy data and detect pathologies by their differences to the learned norm. For this purpose, we use conditional variational autoencoders which learn the reconstruction and encoding distribution of healthy images and also have the ability to integrate certain prior knowledge about the data (condition).

Results Our experiments on different 2D and 3D datasets show that the approach is suitable for the detection of pathologies and deliver reasonable Dice coefficients and AUCs. Also this method can estimate missing correspondences in pathological images and thus can be used as a pre-step to a registration method. Our experiments show improving registration results on pathological data when using this approach.

Conclusions Overall the presented approach is suitable for a rough pathology detection in medical images and can be successfully used as a preprocessing step to other image processing methods.

Keywords Conditional variational autoencoder · Unsupervised pathology detection · Image registration

Introduction

Pathology detection is a crucial but complicated task of medical image computing and analysis. While some applications (e.g., tumor measurement or treatment monitoring) require accurate segmentations of the pathologies, other would only need a detection of the pathology and a rough approximation of its outline, for example to assist the physician in visual inspection or for the retrieval of images with pathologies at specific locations from a database. As another example for such an application, we focus on the registration of images with pathologies. Pathological structures would cause missing correspondences between images and thus lead to wrong registration results in surrounding areas. Hence, it would be helpful to integrate the prior pathology information in the

registration process as an estimation for areas of missing correspondences.

There are many approaches trying to cope with the task of detection or segmentation of pathological structures and especially in recent years machine learning methods gained popularity. The usual approach would be to use supervised machine learning algorithms, which are well specialized at learning the variability of one pathology type and would be often even able to deliver accurate segmentations. For example in [12], a method for the segmentation of brain lesions on the ISLES dataset using random decision forests is developed, where different manually crafted descriptors of the images are classified as pathological or healthy by many trees in a forest.

Other typical supervised learning methods are convolutional neural networks (CNNs). One example for their successful application to pathology segmentation is [4] where a tumor segmentation task is learned using ground-truth annotations of the target structures. A major drawback of such machine learning methods especially neural networks is that they rely on a large annotated training dataset

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from which they can learn a particular task. In this case not only many pathological images are needed, moreover, the ground-truth segmentations of the pathologies are required. In the medical domain, large datasets are hard to access and annotations are very time-consuming, thus usually not available. Another disadvantage is that supervised methods are typically trained on a particular pathology type (e.g., glioblastomas in the brain) and are hence only able to detect this specific pathology type.

For this reason, we pursue a different unsupervised approach based on learning the entire variability of normal healthy tissue and detect pathologies as major differences to the learned norm. For example, a conditional variational autoencoder (CVAE) [5] can be applied to learn a plausible representation of the healthy data variability of a certain domain (e.g., Brain MRI, Lung CT). CVAEs are neural networks which learn low-dimensional encodings by trying to flawlessly reconstruct images and they have been rarely used in the medical image domain by now (acc. to [9]); however, they have the promising ability to model healthy data by mapping it to a low-dimensional latent space with a given prior distribution. CVAEs are especially suitable for this task as they directly learn the latent representation of the input data which is usually very descriptive and tends to focus on the most important features.

A similar idea has been used in [15] for the segmentation of pathological structures in OCT images with the use of GANs. While GANs have a better data generation ability, they are not able to directly map data to its latent representation and need an extra step to do so. A work analogous to ours [14] was also published recently (after initial submission of our work) where a variational autoencoder is used for the unsupervised detection of lesions in Brain CT images. The authors also use the ability of autoencoders to learn the healthy variability of data of a certain domain and thus be able to only reconstruct healthy data; however, they do not make use of the direct latent mapping that autoencoders deliver.

Another more classical unsupervised method is described in [7] and was designed for the detection of brain lesions by modeling the healthy variability with a probabilistic appearance model. Even though unsupervised approaches are usually less accurate, they are theoretically generalizable to all possible anomaly types and have the huge advantage of not needing any annotated training data.

In this work, we inspect the abilities of conditional variational autoencoders for the unsupervised detection of pathologies. Previously, this method was compared to a more conventional dictionary-based approach [19] and showed significantly better results. Here, we explore how the CVAE-based algorithm works on 3D images and its efficiency as an estimator of missing correspondences for a registration method.

Materials and methods

Conditional variational autoencoders

The main idea of an autoencoder is to encode high dimensional data into a low-dimensional latent vector and try to reconstruct the input data as flawlessly as possible by only using its coding. Usually, autoencoders are neural networks that learn the mapping of the data and consist of two parts: An encoder $Q(\mathbf{z}|X)$ which maps the input X to vector $\mathbf{z}$; and a decoder $P(X|\mathbf{z})$ which tries to reconstruct X by only using $\mathbf{z}$. The idea of using autoencoders is that $\mathbf{z}$ is a very descriptive vector of the input data since it delivers sufficient information for the good reconstruction of the input. Thus autoencoders can be successfully used for image representation as in [21] where they manage to learn the representation of cardiac images for cardiac index prediction.

A variational autoencoder (VAE) [5] is a form of autoencoders that assumes a prior distribution of the latent space (typically normal distribution) motivated by the logical distribution of the input data on some particular domain. This is indeed an advantageous feature for our approach, because as mentioned in the further course of this work, we aim to learn a reasonable distribution of the healthy tissue representation of some particular domain.

Furthermore, VAEs can be conditioned (CVAE) [18], by adding an extra prior semantic information about the data, e.g., a label. This way the $\mathbf{z}$ -space distribution is learned per condition and more control over the learned space and reconstruction of images is possible.

CVAE-based pathology detection

For the most part, CVAEs have been used as generative models for computer vision purposes, e.g., [22]. However, in our work, we aim to use them for the unsupervised detection of pathologies. That is, learn a reasonable representation of the healthy tissue variability so that pathological tissue can be detected as a major difference to the learned representation. So by training only on healthy data the VAE would learn a normal distribution in $\mathbf{z}$ -space over the healthy variability of images.

We assume that the latent vectors $\mathbf{z}$ are very descriptive and can actually resemble a good image descriptor. Based on the success of patch-based descriptors [3, 10, 17] and because it would decrease training/testing computational demand, we choose to train the CVAE on small image patches. It is, however, important to integrate location context of the patches, because a healthy looking patch for a certain position might be what pathologies look like at another position, e.g., a patch of bright pixels in a brain MRI resembles healthy tissue on the head border but would be rather pathological if found in the center of the brain. We integrate locality information of the

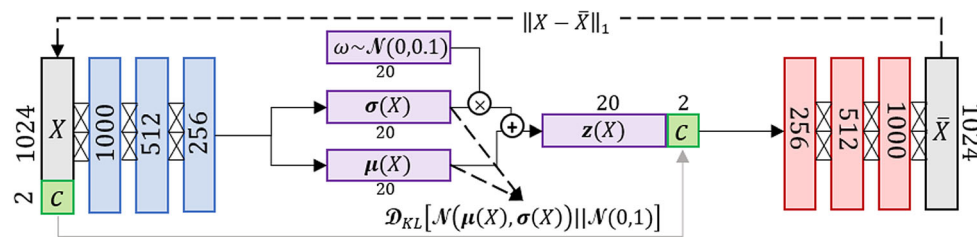


Fig. 1 CVAE architecture for 2D images. Vectorized patch X and reconstruction $\hat{X}$; fully connected encoder (blue) and decoder (red) layers; condition (green); z -space (purple) where $z = \mu + \sigma \times \omega$, with

random $\omega \sim \mathcal{N}(0, 0.1)$. Dashed lines correspond to loss functions: $\mathcal{D}_{\text{KL}}$ —Kullback–Leibler divergence and $\|\cdot\|_1 = l^1$ norm. Numbers denote size of layers

patches by using their continuous positions $c \in [0, 1] \times [0, 1]$ (analog for 3D) relative to the size of the whole image as condition of the CVAE. Thus, the healthy tissue variability is mapped to a normal distribution in z -space per position. Note that in order to ensure that patches on same positions in the different images contain similar structures, an assumption about the rough alignment of the images has to be made, e.g., the images are affinely or rigidly preregistered. Here, even a simpler alignment method was considered to be suitable for not rigidly preregistered data: the images are cropped around the structure of interest using a bounding box, ensuring that the scaling and translation of the structures approximately match relative to the image size.

For the pathology detection in the test phase of the network, two assumptions, further confirmed by our experiments, were made. (1) As the CVAE learns to map healthy tissue to a normal distribution while training, in the test phase healthy patches would also get mapped to the same normal distribution per condition. Pathological patches on the same relative positions, however, differ from healthy tissue drastically, which leads to the assumption that their latent representations are very different to the ones of the healthy patches and lie far outside the normal distribution. Thus, the distance from a z -vector of a pathological patch to the mean of the normal distribution of the healthy training data is large. (2) Another possibility is that by only learning to reconstruct healthy patches the autoencoder reconstructs pathological patches poorly or tries to interpret them as healthy. Hence, big distances between the input image and the reconstruction would occur in the area of the pathology.

Those two assumptions are not strictly disjoint and, as experiments show, should be used together, combined to a third assumption: pathologies have both large reconstruction and z -space distances which corresponds to multiplying both distances.

Architectures

In our work, we developed different CVAE architectures for 2D and 3D data. Here, we show the best ones used for our

experiments. Our CVAE architecture for 2D images is shown in Fig. 1. We choose a simple fully connected architecture containing three hidden layers in the encoder and decoder each.

The more complicated and sophisticated 3D architecture is shown in Fig. 2. Here, we use 3D convolutions in the encoder and 3D deconvolutions (transposed convolutions) in the decoder. The shown architecture is for 3-channel input images but can be analogously used for one channel only.

Note that the dimension of z is crucial—too large z -vectors would contain too much insignificant information and noise; too small z 's might be insufficient to store all the important information about the input. Our experiments showed that 20 and 10 are good dimensions for patches of size 32×32 ; and about 100 for 3D patches sized $32 \times 32 \times 32$.

Another important detail is the choice of the loss functions. While it is rather common to choose the Kullback–Leibler divergence $\mathcal{D}_{\text{KL}}$ as latent space loss, the reconstruction loss may vary. In literature it is usual to take the Sigmoid Cross Entropy Loss when having binary data [5], but the most natural choice for gray-valued data would be l^2 -norm. However, our experience showed that sum of squared differences would lead to learning a very smooth reconstruction with a very small variety between images. Choosing l^1 -norm (sum of absolute differences) leads to more sharp and differentiable results and was thus integrated in our architecture.

In the usual CVAE architecture z would get calculated using the “reparametrization trick” as $z = \mu + \sigma \times \omega$, where $\omega \sim \mathcal{N}(0, 1)$. However, because we enforce $\mathcal{N}(\mu(X), \sigma(X)) \sim \mathcal{N}(0, 1)$ this implies having the same amount of learned information and random noise coming from ω . This is controversial to our detection assumptions and thus we decide to reduce the randomness by sampling ω from distributions with smaller standard deviations (e.g., $\mathcal{N}(0, 0.1)$), which eventually results in having less noisy reconstructions and distributions in z -space.

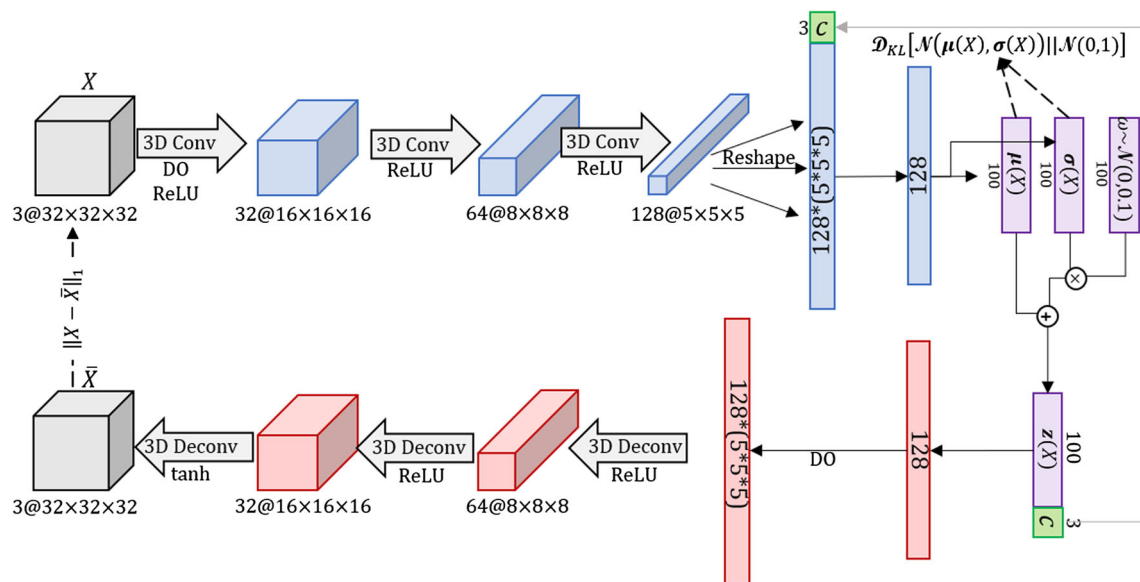


Fig. 2 CVAE architecture for 3D images. 3D patch X (with one or 3 channels) and reconstruction $\hat{X}$; convolutional and fully connected encoder (blue) and decoder (red) layers; condition (green); z -space (purple) where $z = \mu + \sigma \times \omega$, with random $\omega \sim \mathcal{N}(0, 0.1)$. Numbers denote

size of layers (channels). Solid black lines correspond to fully connected operations. DO stays for Dropout; ReLUs are used everywhere if not mentioned otherwise. Loss functions (dashed lines) are same as in Fig. 1

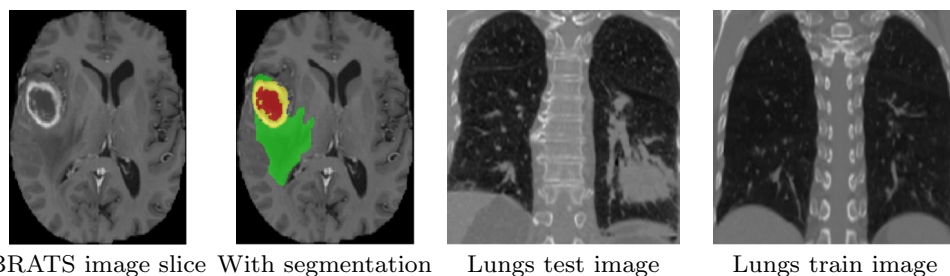


Fig. 3 Examples of the training and testing data for the 2D architecture. From left to right: Slice 80 of the T1c sequence of a BRATS image; ground-truth segmentation of the same BRATS image: enhancing tumor

(yellow), necrotic tissue (red), edema (green). Evaluated only the visible tumor core (red and yellow labels); an example test image of the *Lungs* dataset; an example train image of the *Lungs* dataset

Experiments and results

Data

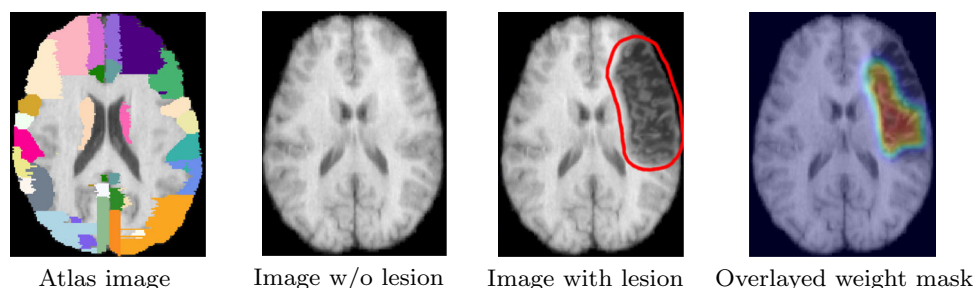
For our experiments, we used three data sets showing different modalities, anatomic regions and pathologies (see Figs. 3, 4 for examples).

BRATS We used 220 Brain MRI images with ground-truth segmentations of high-grade glioblastomas from the BRATS 2015 training challenge dataset [13]. Originally, the images are 3D and contain 4 modalities per subject (T1, T1c, T2, Flair). In the 2D experiments, 2D slices of the T1c images are used because of the good tumor visibility. The whole 3D T1c images or 3-channel multimodal 3D images (T1c, T2 and Flair) are used in the 3D experiments.

A major advantage of this dataset is that the available segmentations allow a selection of non-pathological patches for training and a good evaluation when testing. For the experiments on this dataset, we use a fourfold cross-validation. For training, we choose 150,000 patches not containing pathologies. In the test dataset, we only include subjects containing visible pathologies. When testing on 3D images, we also exclude images that suffer from bad resolution in z -direction (follow-up images). As the skull-stripping of the BRATS images leaves very uneven borders causing high distances in those areas, all distance images are multiplied with the Gaussian smoothed masks of the brain.

Lungs This dataset contains 2D coronal lung CT slices from 46 patients with different pathologies, e.g., lung cancer or fibrosis. There is no suitable ground truth available which

Fig. 4 Examples of LPBA40 dataset. From left to right: atlas image with labels; image without lesions; phantom image with simulated image (red line); image with an overlay of the calculated weight mask (red: no correspondences → blue: present correspondences)



makes picking healthy patches harder and only enables a qualitative evaluation. For training, we thus choose 35 image slices containing no obvious pathologies and extracted 5000 patches from them, and another 11 slices from the remaining images were used for testing.

LPBA40 with phantom lesions The LPBA40 dataset is a publicly available dataset of 40 healthy patients' whole-head MRIs with 56 labeled anatomic regions [16]. The dataset was previously used for evaluation studies of image registration algorithms [6] and selected to show the effects of automatic pathology detection on registration tasks in our experiments. For this purpose, phantom pathological images were simulated using manually segmented stroke lesions extracted from the ISLES challenge data [11]. The lesions are inserted into the LPBA40 images as follows:

$$I^{\text{Pha}} = \text{HE}(T(I^L)) \cdot M^L + I \cdot (1 - M^L) \quad (1)$$

where I^L denotes a lesion image from the ISLES dataset, I is a healthy LPBA40 image and M^L is a smoothed segmentation mask of the lesion. $\text{HE}(\cdot)$ denotes a histogram equalization to match the intensities of both images, and $T(\cdot)$ is an affine transformation required to register both images to each other. In our experiments, we used only one lesion simulated onto all LPBA40 images resulting in 40 healthy and 40 corresponding pathological images with known ground-truth segmentation of the pathologies (see Fig. 4 for examples).

Here, we also apply a fourfold cross-validation over the patients always using 30 healthy patients images for training (6000 patches extracted) and the remaining 10 with simulated lesions for testing, making sure that the train and test dataset never contain the same patients.

As it is important for the images to be roughly aligned to each other, they all have been cut using a bounding box around the structures (brain or lungs). For this reason, the sizes of the images vary, but the relative positions of the structures match approximately. Another important detail is that all extracted training patches are sampled on random positions to avoid learning only discrete positions. Our experience showed that patches sized 32×32 (or $32 \times 32 \times 32$ for 3D) deliver best results; hence, this was the used patch size in all experiments. From the test images, we sample

same-sized patches densely with large overlaps ($> 50\%$ of the patch size).

Experimental setup

Pathology detection with the CVAE architecture

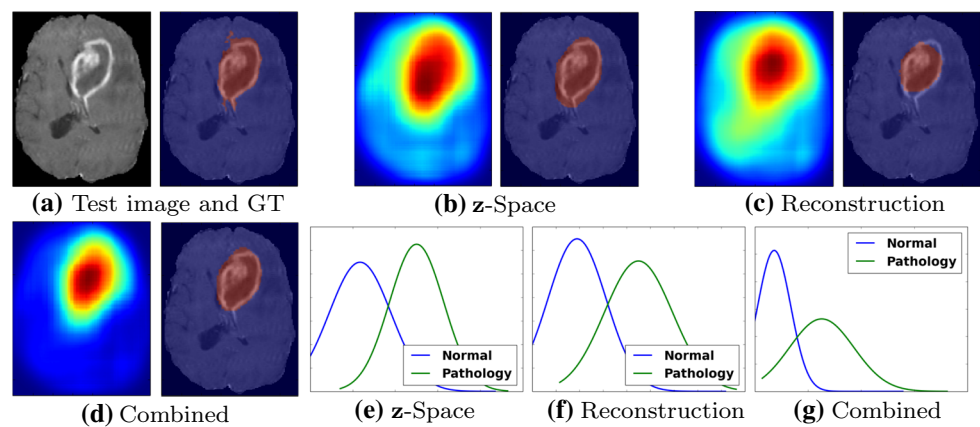
For our experiments, we applied the 2D (Fig. 1) and 3D (Fig. 2) CVAE-based approaches. We investigated all three detection assumptions: reconstruction distances, $\mathbf{z}$ -space distances and combined distances. The distances were calculated patch-wise, resulting in a single value per patch. The reconstruction distances correspond to the l^2 -norm of the pixelwise difference of the original and CVAE-reconstructed patches; the $\mathbf{z}$ -space distances are computed using Euclidean distances of the $\mathbf{z}$ -representation of an input patch and the mean of the $\mathbf{z}$'s of the healthy training data; and the combined distances are the multiplication of the both previously mentioned. The patch distances are then averaged over the overlapping patches, which leads to smoother results than using pixelwise distances from non-overlapping patches.

To evaluate the differentiation ability of those distance images, we execute a global ROC analysis over all images in a dataset on the possible lower thresholds (as a percentage of the maximum value of the distance images) and calculate the area under curve (AUC). Sensitivity and specificity values are calculated on the Youden index ($\max(\text{sensitivity} + \text{specificity} - 1)$) of the ROC. We also used this index to determine a global threshold for the pathology segmentation. The resulting segmentations are then evaluated with Dice coefficient.

Those metrics were used in all experiments excluding the *Lungs* dataset with no available ground truth. However, the number of test images is small enough to enable a reasonable visual evaluation in this case. For the experiments on the 2D and 1-channel 3D BRATS, we only evaluate the tumor core (Fig. 1) since this is the primarily visible tumor part on the T1c modality.

For the 3-channel 3D BRATS experiments, we separately evaluate every modality according to the visible tumor parts in it (T2 and Flair include the tumor edema). We also consider a combination of the reconstruction distance maps of all

Fig. 5 2D BRATS segmentation results. An example image with its ground-truth segmentation. **a** The resulting distance images (growing values: blue $\rightarrow$ red) and segmentations of our method with the three detection assumptions. **b–d** Distributions of distances of pathological (green) and healthy (blue) data over all test images **e–g**



three modalities and refer to it as multimodal distance (MM), where $MM = \alpha D_{T1c} + \beta D_{T2} + \gamma D_{Flair}$ with $\alpha + \beta + \gamma = 1$ and D_{T1c} , D_{T2} , D_{Flair} are the reconstruction distance images for each modality respectively. Our experience showed that $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.3$ lead to good results. The multimodal distance is evaluated regarding all tumor labels.

For the LPBA40 dataset, we also evaluate the pathology detection ability using the lesion masks as ground truth. To underline the importance of using a condition in the variational autoencoder architecture (c -feature in Figs. 1 and 2), we run an additional experiment, using the exactly same setup on the LPBA40 data, except that the conditional features c are left out in the architecture of the neural network.

For the 2D architecture applied on BRATS, we used $dim(\mathbf{z}) = 20$ and for the Lungs $dim(\mathbf{z}) = 10$. The 3D architecture naturally requires higher dimension of $\mathbf{z}$, and we used $dim(\mathbf{z}) = 100$ for all 3D experiments.

Pathology detection for nonlinear image registration

In this experiment, we examine the benefit of our pathology detection method in atlas-to-patient registration scenarios assuming that a healthy atlas image is registered to a patient image containing pathological structures. Therefore, one subject of the LPBA40 data is selected as atlas and registered to the images of 39 remaining subjects using a variational nonlinear registration method [20]. The registration method was chosen because it performs among the best algorithms on the LPBA40 dataset and is freely available [2]. Normalized local cross correlation (NCC) is used as distance measure during the registration combined with curvature regularization; all parameters were chosen according to [2].

Three registration tasks are performed: (1) using the original (healthy) LPBA40 images, (2) using the phantom pathological images, and (3) using the phantom pathological images combined with a weight mask. To compute this weight mask for each subject image, the combined reconstruction and $\mathbf{z}$ -space distances D from the CVAE

(see “CVAE-based pathology detection” section) are thresholded (see “Pathology detection with the CVAE architecture” section), resulting into a segmentation S . Then, S is Gaussian smoothed with $\sigma = 3$ ($gauss(S)$). The weight mask $w(\mathbf{x}) \in [0, 1]$ is the inverted version of $\bar{w} = \max(resc(D), gauss(S))$, where $resc(\cdot)$ denotes rescaling the images to the interval $[0, 1]$ and $\max(\cdot)$ is a voxel-wise maximum function. The final inverting of the values ensures that $w(\mathbf{x}) \in [0, 1]$ with $w(\mathbf{x}) = 0$ for regions clearly detected as pathology. The resulting mask locally weights the distance measure and thus suppresses the influence of pathological regions. To evaluate the registration performance, Dice indices between the registered atlas labels and the subject-specific ground-truth labels are computed and averaged over all 56 labels. Also we compute the standard deviations of the Jacobians to inspect the smoothness of the atlas-to-patient transformations [20].

Results

Pathology detection on 2D images

The results on the 2D BRATS data are shown in Table 1 and Fig. 5. Clearly, the ROC analysis shows that the method is able to establish differentiation between healthy and pathological tissue ($AUC \gg 0.5$). This is also visible in the distribution plots of the distances of healthy and pathological patches in Fig. 5, where pathologies clearly tend to deliver bigger distances. Best results in terms of all evaluation metrics are achieved when using the combination of reconstruction and $\mathbf{z}$ -space distances resulting off the CVAE-method. This can also be seen in the example image in Fig. 5 where the oversegmentation b) of the $\mathbf{z}$ -space and the undersegmentation c) of the reconstruction are combined to a nearly optimal segmentation d) of the pathology.

The achieved mean Dice of 0.55 is a segmentation result comparable to the results from [13] where it would make the 6. place, but not to newer approaches like [4], where CNNs

Table 1 2D detection results for BRATS

	Reconstr	z-space	Combined	anoGAN
Dice	0.49 ± 0.26*	0.51 ± 0.26*	0.55 ± 0.27	0.51 ± 0.28*
Sens	0.89	0.88	0.91	0.90
Spec	0.81	0.85	0.86	0.81
AUC	0.94	0.92	0.95	0.93

Shown are from left to right: CVAE reconstruction, **z**-space and combined distances; anomaly score of anoGAN from [15]. Evaluation metrics are mean over all images and folds: Area under curve (AUC) from ROC analysis over all subjects, sensitivity and specificity at Youden index and Dice coefficient of segmentation with globally ROC-picked threshold.

*Corresponds to statistical significance ($p < 0.001$) in a one-sided t test compared to combined

are used (note that the BRATS challenge uses 3D multimodal data). Still, all of those approaches are supervised and trained for the explicit segmentation of glioblastomas. Our method, however, establishes an unsupervised detection of different pathologies that are not necessarily considered by the used ground truth, but would still cause missing correspondences. Furthermore, 2D layers with no visible pathologies would often contain parts of overflowing 3D segmentations. Thus, right detection by our method would result in bad Dice coefficients.

For comparison, we implemented the so-called anoGAN method from [15]. AnoGAN is a GAN-based method, also following the idea of learning the representation of healthy images. Similarly to the approach in this work, the authors combine a **z**-space distance and an image reconstruction distance of the images in test phase, defining an anomaly score to differentiate between pathologies and healthy tissue. However, GANs only learn the direct mapping $G(\mathbf{z}) = \tilde{x}$ from the latent **z** to a generated image $\tilde{x}$, and not the other way around. Thus, to find the latent mapping of a test image x , the **z**-space is sampled until $G(\mathbf{z}) \approx x$ by solving an additional optimization problem. Hence, the inference of the model is significantly slowed down. Since we apply anoGAN patch-wise on densely sampled overlapping patches, analogously to the CVAE experiments (using the per patch anomaly score to generate distance images), the significantly longer test phase per patch (about ≈ 15 times slower in our experiments) leads to unfeasible computational times for a whole image.

Still, the exact same experimental setup for the 2D BRATS images is recreated using anoGAN and its results can be seen in Table 1. Note that the approach was originally developed for OCT data and we can not ensure that our architecture and training procedures are identical to [15]. In our observations, however, anoGAN tends to better reconstruct the pathological structures, possibly due to the lack of positional conditions, the lack of constraint on the **z**-space distribution, and the indirect optimization-based **z**-space mapping. Therefore, anoGAN shows significantly lower Dice scores and seems to be less suitable for the task presented here.

Even though the *Lungs* data set is fundamentally different, the visual evaluation showed that the CVAE-method still has

the ability to detect present pathologies (Fig. 6 big distances correspond to pathological structures). Interestingly, **z**-space distances here seem to rather correspond to edges than pathological structures since pathologies typically have the same intensity values as healthy tissue. Still, combining them with the reconstruction distances performs well.

Pathology detection on 3D images

The 3D pathology detection results are shown in Table 2. The results point out that the 3D detection task is much harder than 2D for the CVAE-approach. The results on both the LPBA40 and BRATS 3D datasets are comparable. However, when training BRATS on only one modality, the values are not convincing. Thus, it is clearly an enrichment to use multiple modalities—even the distances on the T1c modality deliver higher Dice values when training on all modalities than the T1c distances on a single modality. Best results regarding the segmentation quality based on Dice for the BRATS data are achieved when using the multimodal detection. Here, a Dice value of 0.5 is achieved and as a one-sided t test shows this is significantly better than the other approaches. Interestingly, higher Dice values not always correspond to higher AUC values. For example, the 3-channel T1c approach has a higher AUC than the MM approach. This is explained with the much higher specificity based on the large amount of true negative segmented pixels (typical for undersegmentation as the major part of the background gets classified correctly). However, true negative pixels are not considered when calculating Dice coefficient. In Fig. 7, the different distance images for a single BRATS slice are shown. All of them tend to have large values at the pathological spot and the combination of them ensures best differentiation of the healthy and pathological tissue.

The LPBA40 results shown in Table 2 reach a Dice of 0.49. However, when leaving out the conditional features in the VAE architecture, a significant performance drop can be noticed: Dice coefficient drops to 0.42 ± 0.13 and AUC to 0.88 using the exact same setup. This underlines the intuition that conditional features are overall meaningful, as they give additional prior information about the data and thus a smaller dimension of **z** is necessary. When leaving this explicit infor-

Fig. 6 2D Lungs distance images. Two example images with their corresponding distance images (growing values: blue → red) with the three detection assumptions

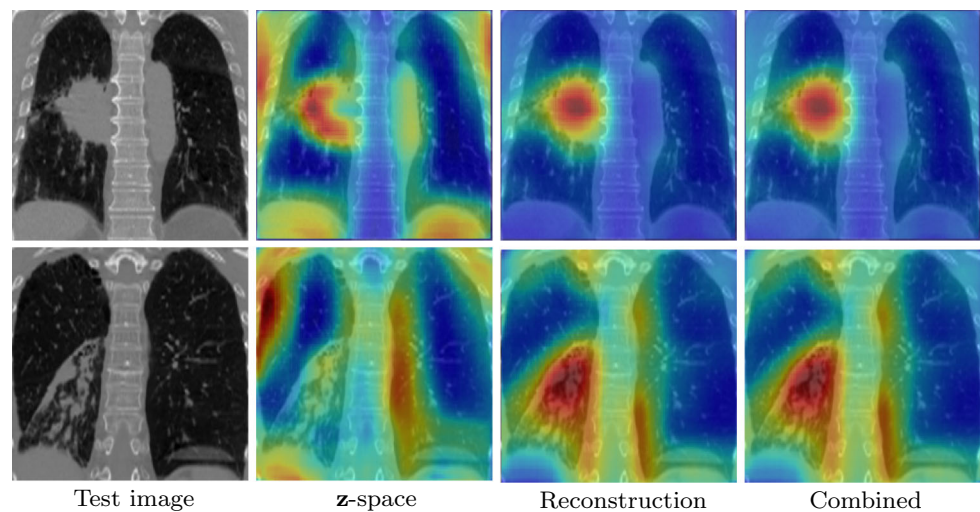


Table 2 3D detection results for BRATS and LPBA40

	BRATS	BRATS 3-channel				LPBA40
	T1c only	T1c	T2	Flair	MM	
Dice	0.32 ± 0.24	0.47 ± 0.17	0.29 ± 0.22	0.40 ± 0.17	0.50 ± 0.18*	0.49 ± 0.11
Sens	0.89	0.92	0.93	0.92	0.91	0.88
Spec	0.86	0.87	0.80	0.80	0.81	0.86
AUC	0.95	0.96	0.93	0.93	0.94	0.93

From left to right: the combined distances for BRATS one channel; BRATS multichannel T1c, T2, Flair (separately) and multimodal (MM); LPBA40. Evaluation metrics are mean over all images and folds: Area under curve (AUC) from ROC analysis over all subjects, sensitivity and specificity at Youden index and Dice coefficient of segmentation with globally ROC-picked threshold

*Corresponds to statistical significance ($p < 0.05$) in a one-sided t test compared to all other BRATS experiments

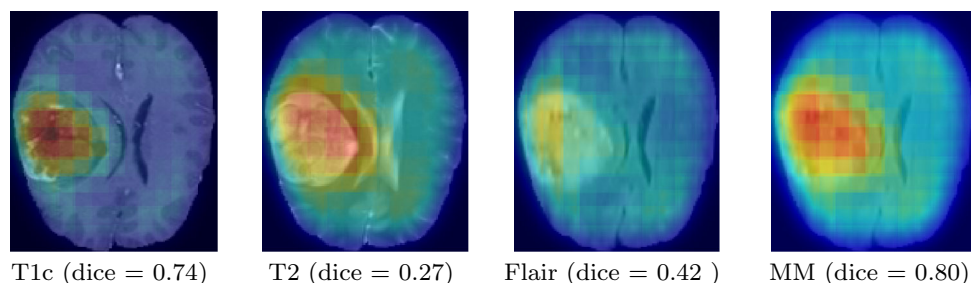


Fig. 7 3D BRATS Multimodal distance images. Different sequences of an image with overlaid distances. From left to right: T1c with overlaid T1c distance; T2 with overlaid T2 distance; Flair with overlaid Flair distance; Flair with overlaid multimodal distance (growing values: blue → red)

mation out, it needs to be implicitly learned by the VAE and stored in larger z -vectors, which also implies a longer training process. Furthermore, location-related conditions are necessary in this particular case, because healthy tissue is only detectable in the context of its localization (e.g., very dark pixels are detected as healthy in the ventricles, but pathological in the brain tissue), which also explains the better performance when using conditions.

Pathology detection for image registration

The image registration results are shown in Table 3. The given Dice values, averaged over all labels and subjects, show that pathological structures induce a decreased registration performance. However, including the distance weight mask into the registration process significantly ($p < 0.001$) improves the results since the influence of the non-corresponding areas is suppressed. Additionally, pathologies influence the smoothness of the resulting transformation, because miss-

Table 3 Image registration results

	No lesion	With phantom lesion	
		No weights	With weights
Dice	0.74 ± 0.01	0.72 ± 0.01	0.73 ± 0.02
Std. Jac.	0.47	0.55	0.37

The Dice coefficients and standard deviation of the Jacobian for all three registration experiments. Dice coefficients are computed between registered atlas labels and subject-specific ground-truth labels and averaged over all 56 labels and all subjects. The standard deviation of the Jacobian represents the smoothness of the resulting atlas-to-patient transformation [20] and is computed on the voxel-grid and averaged over all subjects

ing correspondences result in implausible distortions of the displacement field and consequently in higher displacement gradients. This can be observed in Fig. 8 where large distortions of the displacement field appear in the pathological areas when no weight mask is used. However, when incorporating the weights mask into the registration process we observe a much smoother displacement field in the region of the lesion. To evaluate this quantitatively, the standard deviation of the Jacobian is used to measure the smoothness of the computed transformation (as proposed in [20]). It can be observed that usage of a weight mask leads to a much smoother transformation, given by a significantly smaller Jacobian standard deviation ($p < 0.001$). This can also be noted in Fig. 8 where the standard deviation of the Jacobian when not using a weight mask is as expected much larger than in the contrary case. The results show that integrated knowledge about the presence and location of pathologies strongly improves registration performance.

Discussion

In this work, we presented an unsupervised machine learning approach for pathology detection in medical images. The leading idea of the presented method is to enable rough detection of pathological or abnormal areas in images without the

use of ground-truth-annotated data. The development of such unsupervised methods is important when it comes to medical images as the generation of ground-truth annotations by experts is very time-consuming and thus ground-truth data is usually not available in the large amounts needed for training a neural network. For our approach, we make the assumption that pathologies can be recognized as regions that do not look like the normal tissue specific for a certain location. Thus, it is not necessary to learn the variability of a certain pathology type from many segmentations, but to learn the natural variability of healthy data of a certain type and later recognize pathologies as regions that are very different to the learned norm. We argue that a conditional variational autoencoder (CVAE) is a suitable tool to learn a good representation of the healthy data because autoencoders are able to learn good low-dimensional representations $\mathbf{z}$ of their input by fulfilling the auxiliary task to reconstruct it. Also a CVAE has the advantage to learn a distribution over the $\mathbf{z}$ -space per certain condition.

We train the CVAE on healthy patch data and use the relative positions of the patches as condition and thus integrate locality into the approach. We assume that this way the CVAE learns to reconstruct healthy data well and would fail at reconstructing pathological test patches. Another assumption is that pathological patches would get mapped far away from the learned distribution in $\mathbf{z}$ -space. Our experiments show that both assumptions are plausible and a combination of both works best.

We test the detection abilities of our approach by first applying a simple fully connected CVAE architecture to two very different 2D datasets: Brain MRIs with tumors and Thorax CTs with diverse pathologies. In those first experiments, the CVAE-method is shown to be clearly suitable for the detection of pathologies and generalizable to many different image and pathology types. Surely, the quantitative results are not comparable to new supervised CNN-based segmentation methods; they are however promising for an unsupervised approach and to establish a rough pathology detection. In a second step, a more sophisticated architecture

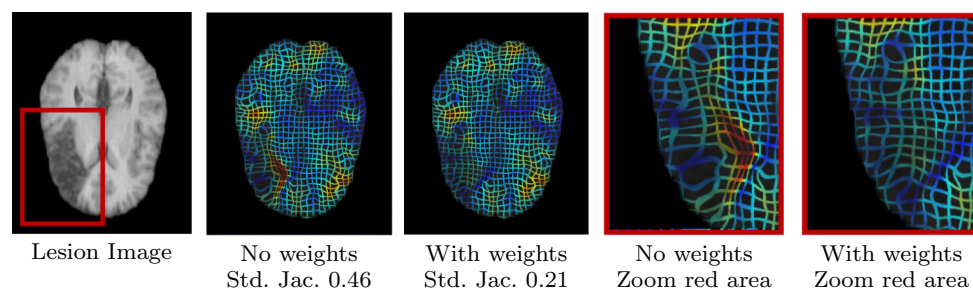


Fig. 8 Visualization of the magnitude of the registration displacement field (atlas to a lesion reference image). Magnitude increases blue to red. From left to right: A reference image with a pathological (lesion) area

(marked red); Overlaid registration displacement fields when using no weight mask and when using a weight mask with given standard deviations of their Jacobian; zoomed into the area of the red rectangle

with 3D convolutions is applied for 3D data. The tests on two brain MRI datasets (BRATS with glioblastomas and LPBA40 with phantom lesions) show that learning the 3D variability of the healthy tissue is clearly a much more complex task as the quantitative values drop. Especially, the BRATS 3D dataset seems to be a challenge, but reasonable results could be achieved by using a combination of multiple modalities of the BRATS images.

The LPBA dataset seems to be less complex as it delivers comparable results without the availability of multiple modalities of the images. In this study, however, a simple segmentation using a global threshold is applied on the computed distance maps to show the principal capability of our approach. Dice coefficients will increase when more sophisticated segmentation methods are applied, e.g., region growing or adaptive hysteresis thresholding.

A major motivation behind this work is that pathologies are structures that hinder correspondence finding in an image registration process. Therefore, in our last experiment, we investigate whether our detection method can be used as a preprocessing step to a variational registration method. The distance maps resulting from the CVAE-based detection are used as weight masks of the distance measure of the registration method to suppress the influence of pathological regions. We show that this way the variational registration approach performs significantly better compared to not using any weight masks. This experiment proves that even very rough pathology detection can be meaningful to further image processing methods and confirms the adequate pathology detection of the method. However, much space for experiments on creating the weight mask from the distance images is left and surely even better results are achievable.

Overall, we show that CVAEs are able to learn suitable representations of data with application to pathology detection. Especially on 2D images the detection approach performs well. On 3D images, we are still able to enrich the representation learned by the CVAE by using multimodal data. However, when the data gets more complicated (3D data for example), the representation and reconstruction abilities of CVAEs seem to drop. This might be related to the chosen CVAE architecture for 3D images and will be investigated in future work. For example, the conditional information is not used in the first layers of the encoder, and the size and number of convolutional layers can be varied. It is further known that VAEs tend to reconstruct the input data rather blurry compared to methods like generative adversarial networks (GANs) or conditional GANs. A recent work suggests a combination of (C)VAE and GAN that allows direct mapping to latent space while preserving the reconstruction quality of GANs [1,8]. This can be another promising concept to be investigated for our application.

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Compliance with ethical standards

Conflict of interest Hristina Uzunova, Sandra Schultz, Heinz Handels and Jan Ehrhardt declare that they have no conflict of interest.

Human and animal rights This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent For this type of study, formal consent is not required.

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